Experiment 10: Applying Performance Optimization Techniques in Zoho Creator

Aim

To implement performance optimization techniques, including field indexing, subform structuring, and Deluge query optimization, in the Zoho Creator Job Application Management application to improve record retrieval, simplify data management, and enhance application efficiency.

Objectives

- To identify frequently searched fields in the recruitment application.
- To examine field indexing options for improving record retrieval.
- To understand how subforms can organize related interview details.
- To optimize Deluge scripts by using specific search criteria and avoiding unnecessary database operations.
- To test forms, reports, charts, and workflows after optimization.

Overview

Performance optimization improves the efficiency of applications that manage large amounts of data. In the Job Application Management application, candidate records, application statuses, interview details, and department-wise reports must be retrieved and processed efficiently.

The following techniques can be used:

1. Field Indexing: Helps speed up searches on supported fields when configured appropriately.
2. Subform Structuring: Groups related information, such as multiple interview rounds, within a parent application form.
3. Deluge Query Optimization: Retrieves only the required records and avoids unnecessary database operations.
4. Performance Testing: Checks whether forms, reports, and workflows function correctly after changes.

Procedure

Step 1: Identifying and Applying Field Indexing

1. Open the Zoho Creator Developer workspace.
2. Open the Job Application Management application.
3. Open Job_Application_Form in the Form Builder.
4. Identify frequently searched fields, such as Candidate Email, Application Status, and Department Applied For.
5. Open the field properties and check whether indexing options are available.
6. Configure indexing only if the option is supported.
7. Do not enable the Unique property unless duplicate values must be prevented.
8. Save the changes and verify that the form works correctly.

Step 2: Structuring and Optimizing Subforms

1. Open Job_Application_Form in the Form Builder.
2. Review the existing fields related to candidate interviews.
3. Identify related information that could be grouped into a subform.
4. For example, an Interview_Details subform could contain Interview Round, Interview Date, Interviewer Name, and Interview Feedback.
5. Add the subform only if it suits the existing application structure.
6. Keep only the necessary fields to make data entry simple.
7. Save the form and test whether the interview details are recorded correctly.

Step 3: Writing Optimized Deluge Scripts

1. Open the existing workflows or functions in the application.
2. Review the scripts used to retrieve candidate records.
3. Replace broad searches with specific criteria wherever possible.

Example of a broad query:

```
candidates = Job_Application_Form[ID != 0];
```

Example of a filtered query:

```
candidates = Job_Application_Form[Application_Status == "Applied"];
```

4. Use the exact field link names and field values configured in the application.
5. Avoid repeated database searches inside for each loops when the same data can be retrieved once.
6. Retrieve only the records required for the task.
7. Review functions such as Generate_Offer_Letter_Function for unnecessary record lookups.
8. Save and test the scripts to ensure that existing recruitment workflows continue to work correctly.

Step 4: Testing and Monitoring Performance

1. Open Candidate_Application_Report and verify that candidate records are displayed correctly.
2. Open Applications_Spreadsheet and check the screening and interview details.
3. Open Department_Applications_Chart and verify that the department-wise application statistics are displayed correctly.
4. Submit a sample application and confirm that the record is saved.
5. Test the relevant workflows and functions.
6. Review available execution logs for errors or unnecessary operations.
7. Compare performance before and after optimization if measurements are available.
8. Record the results and note any features that could not be configured.

Expected Output

- Candidate records are retrieved using appropriate search criteria.
- Related interview information can be organized using a subform, if implemented.
- Deluge scripts avoid unnecessary database operations.
- Recruitment forms, reports, charts, and workflows function correctly.

Result

Performance optimization techniques were studied and applied where supported in the Zoho Creator Job Application Management application. Frequently searched fields were reviewed, subform structuring was evaluated for interview details, and Deluge queries were refined to retrieve relevant candidate records. The forms, reports, charts, and workflows were tested to verify that the existing recruitment functionality remained intact.